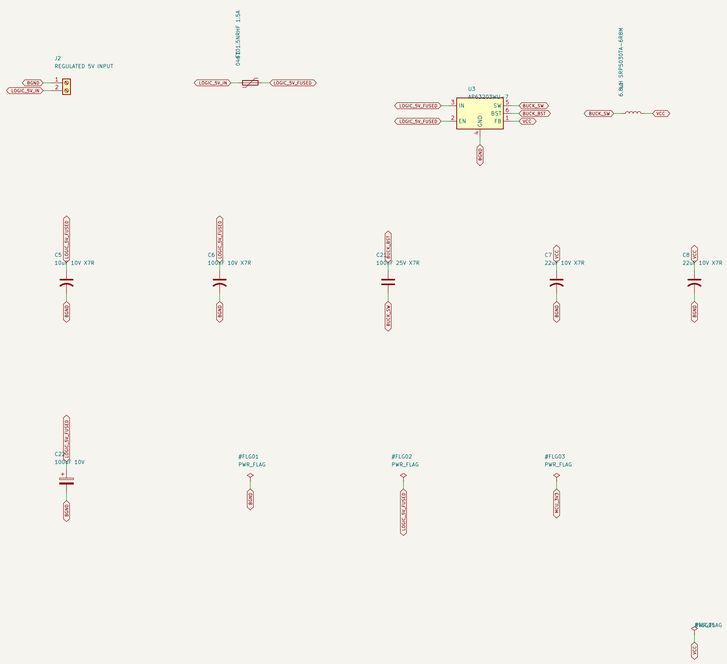
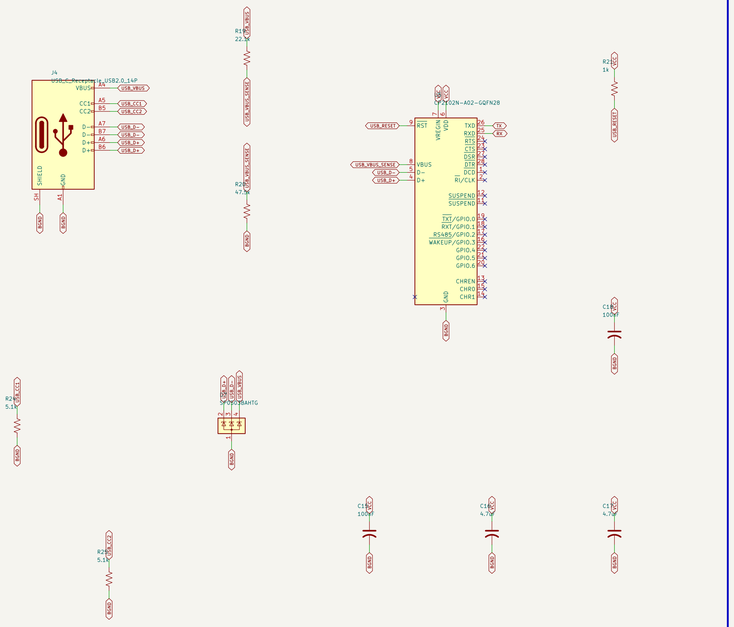
## 01 POWER / 5 V TO 3.3 V

J2 regulated 5 V only. Buck + local reservoirs. Not battery charging.



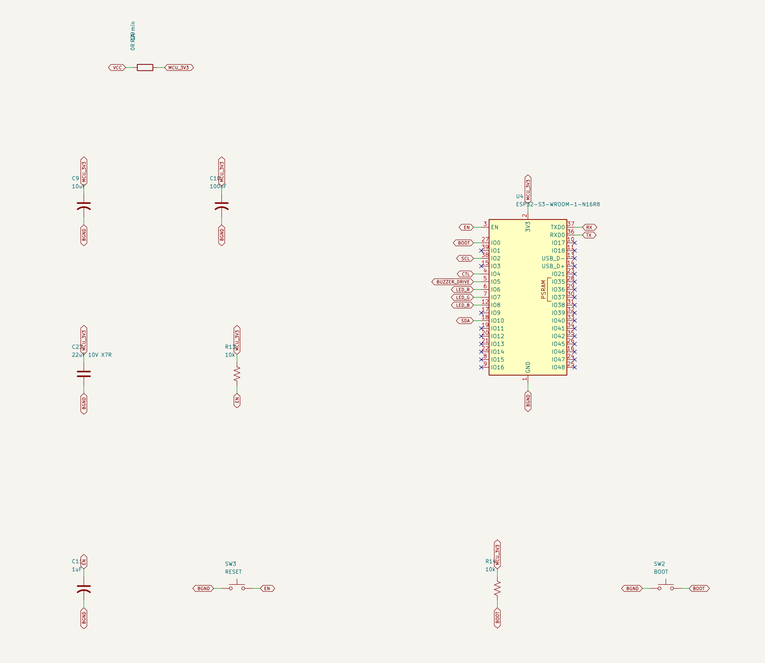

## 02 USB / UART PROGRAMMING

USB is data only; J2 powers the board. Hold BOOT, pulse RESET.



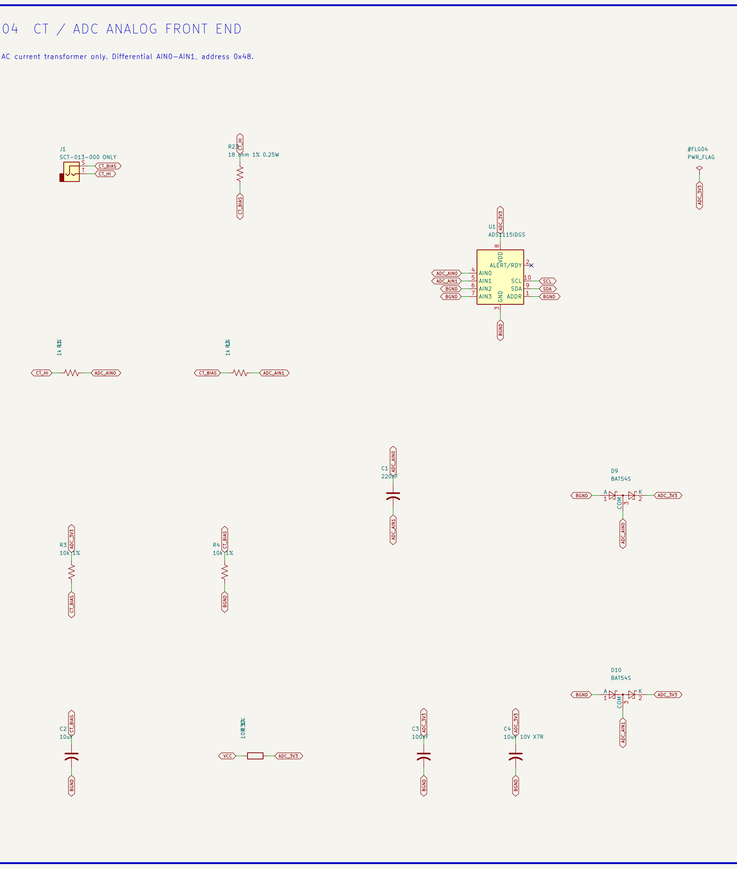

## 03 ESP32-S3 / BOOT / RESET

ESP32-S3-WROOM-1-N16R8. Module RF keepout on every copper layer.



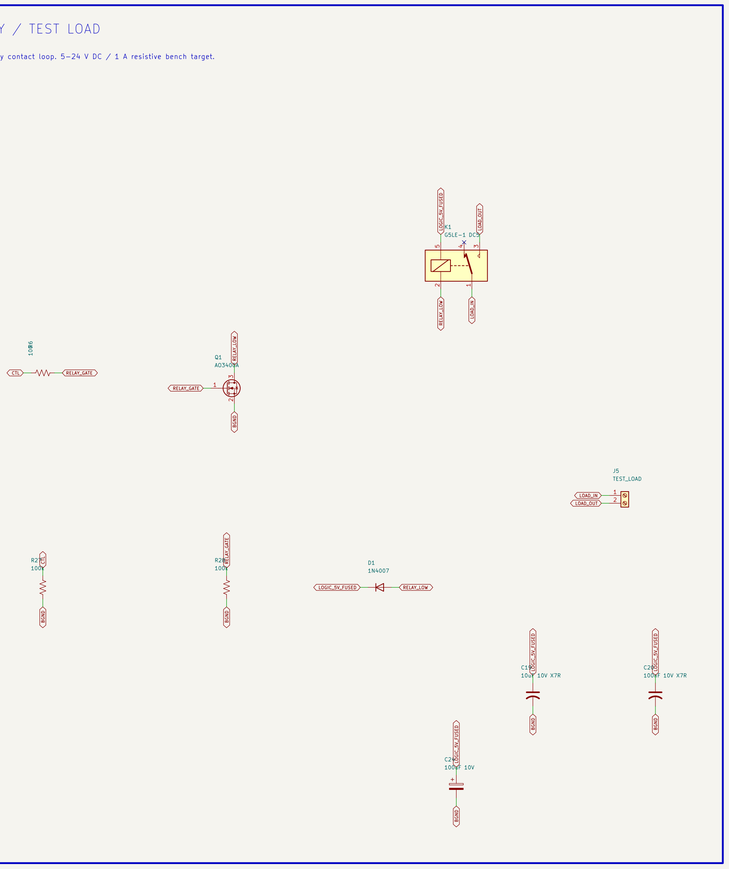

## 04 CT / ADC ANALOG FRONT END

AC current transformer only. Differential AIN0–AIN1, address 0x48.



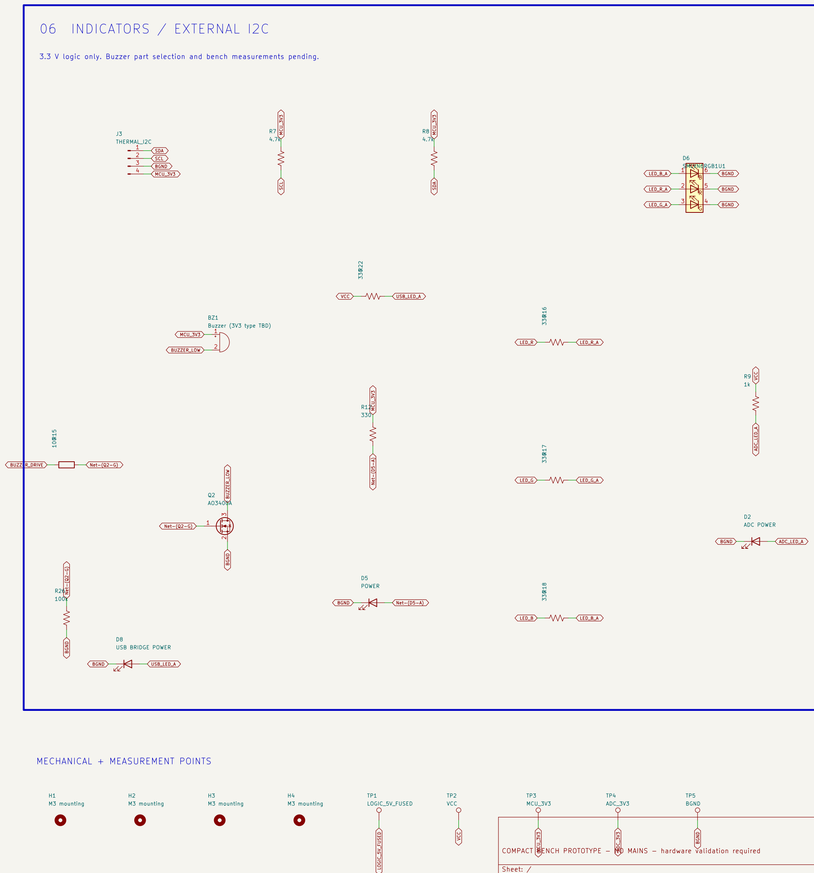

## 05 RELAY / TEST LOAD

Normally-open dry contact loop. 5–24 V DC / 1 A resistive bench target.

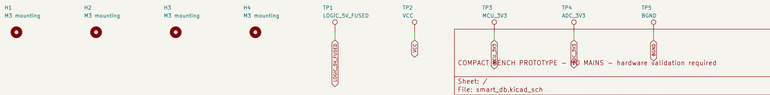



## 06 INDICATORS / EXTERNAL I2C

3.3 V logic only. Buzzer part selection and bench measurements pending.




MECHANICAL + MEASUREMENT POINTS




COMPACT BENCH PROTOTYPE — NO MAINS — hardware validation required

Sheet: /
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**Title: SmartDB v2 — low-voltage bench revision**

| | | |
|---|---|---|
| Size: A1 | Date: 2026-10-08 | **Rev: 2-compact** |
| KiCad E.D.A. 10.0.6 | | Id: 1/1 |